Final Deliverable Report

SI 495 - Information Analysis Capstone 2

Group 3

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Ann Arbor Hands-On Museum

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Project Overview

Our client is the Ann Arbor Hands-On Museum (AAHOM), a nonprofit organization whose mission is to create moments of discovery that inspire curiosity for Science, Technology, Engineering, Art, and Math (STEAM)[1]. They serve over 300,000 learners in their Kerrytown and Leslie Nature Center locations, providing field trips, outreach programs, and distance-learning opportunities across Michigan and beyond[2]. Our specific clients are Charlie Reischl, Assistant Director of Advancement, and Ari Morris, Director of Exhibits and IT. Before we started this project, the museum had lots of scattered data that they didn't know how to consolidate, visualize, or interpret. This project helped the museum better track its attendance, revenue, and donations from visitors. We produced dashboards in Apache Superset, a software application for data exploration and data visualization. Superset is mapped to the museum's SQL based Customer Relationship Management (CRM) software (Versai) to interpret attendance and revenue data to inform the organization's leadership to make strategic decisions. More specifically, the client wanted us to build out attendance, donor, and membership dashboards, which will all provide a holistic view of the communities served, helping leadership identify underserved populations and expand access to their programs.

Deliverable Description

The primary deliverables for this project are three dashboards and a training guide. The first dashboard is the attendance dashboard, which includes visualizations that track attendance across the museum's different programs over time, showing cumulative attendance and instances for each program type compared to 52 weeks ago. There are also visualizations showing the museum's general attendance

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[1] "Ann Arbor Hands-On Museum and Leslie Science and Nature Center." *Ann Arbor Hands-On Museum and Leslie Science & Nature Center*, discoverscienceandnature.org.

[2] *Ann Arbor Hands-On Museum and Leslie Science and Nature Center: Creating Moments of Discovery | Songfoundation*. song.foundation/ann-arbor-hands-on-museum-and-leslie-science-nature-center-creating-moments-of-discovery.

segmented by categories such as “Member vs. Non-Member attendance” and “Family vs. School attendance.” The attendance dashboard will help program managers understand which programs are happening on a given day and which have been the most popular. The second dashboard is the membership dashboard, which displays visualizations related to the museum’s members across the three membership tiers. We made visualizations of membership sign-ups in the museum versus online from this year MTD compared to last year MTD. This dashboard will help museum staff understand who they can better target to increase membership, and which method is more effective (in-person vs. online). The last dashboard is our donor dashboard that analyzes donor behavior to understand giving patterns, donor retention, and engagement levels. We segmented donors by the total amount donated, and from there mapped each individual donor's total amount, number of donations, and donation recency. We built this dashboard to analyze monthly donation metrics, contrasting the total number and monetary value of gifts from organizations against those from households. The donor dashboard will help the museum improve fundraising efforts and growth; more specifically, understand which donors are likely to donate but haven’t donated in a while. All these dashboards will provide value and help museum staff better understand attendance, membership, and donor trends, identify outreach opportunities, and make informed strategic decisions. Our client can use and maintain our dashboards independently after the project ends because the dashboards will automatically update with the most recent data. Additionally, we will be providing them with an informative handoff session and robust documentation that will outline what they can do if they would like to change certain parts of the charts in the future, which will be described in more detail in a later section. Overall, these materials ensure that museum staff can independently update, maintain, and scale the dashboards without reliance on the student team.

Results & Analysis

As described earlier, we developed three main dashboards (attendance, membership, and donor). We refined our visualizations throughout the semester multiple times, applying the feedback that the client provided us constantly. In our client meetings, we asked them how they would use our dashboards and the specific charts within them as well.

Our workflow included first finding relevant datasets to obtain the visualizations we wanted. This meant uploading data from Versai, joining tables within Superset, or extracting certain columns from existing tables. We would then create SQL queries within the built-in SQL lab on Superset, analyze the data, create charts, and make variations as needed. In this section, we would also like to highlight a few of the many visualizations we developed to give a brief look into our work. Below, we will show one visualization from each dashboard.

First, we tracked the cumulative museum attendance compared to 52 weeks ago, as shown below for the attendance dashboard. We took in feedback from the instructional team to include cumulative data as opposed to individual months' data. This helped us develop more holistic interpretations and account for small time periods when attendance could be low without affecting insights for the leadership. In addition to tracking general cumulative attendance, we also tracked cumulative attendance for each program type that the museum offers (i.e field trips, distance learning, public programs, etc). For the sake of clarity, we included a couple program charts in the appendix.

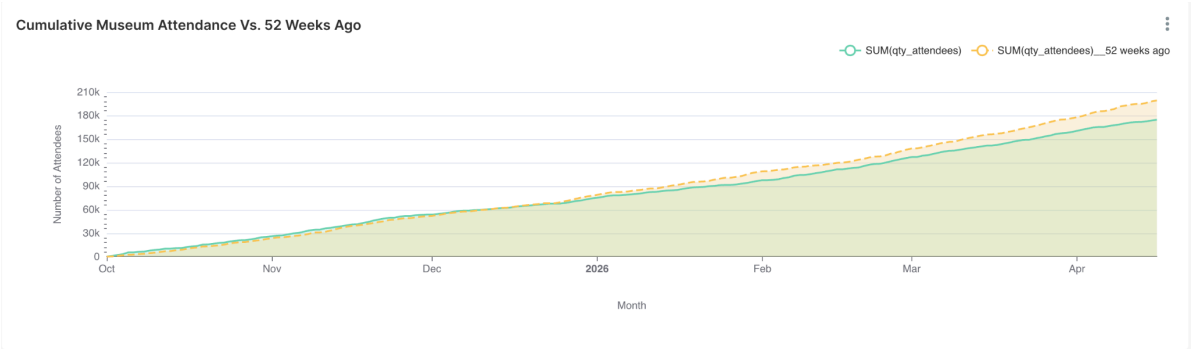
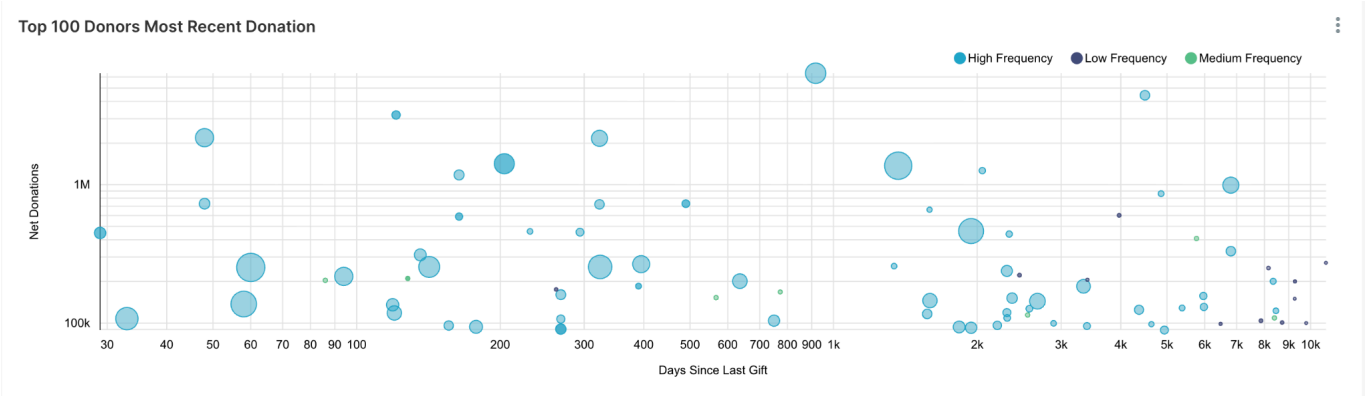


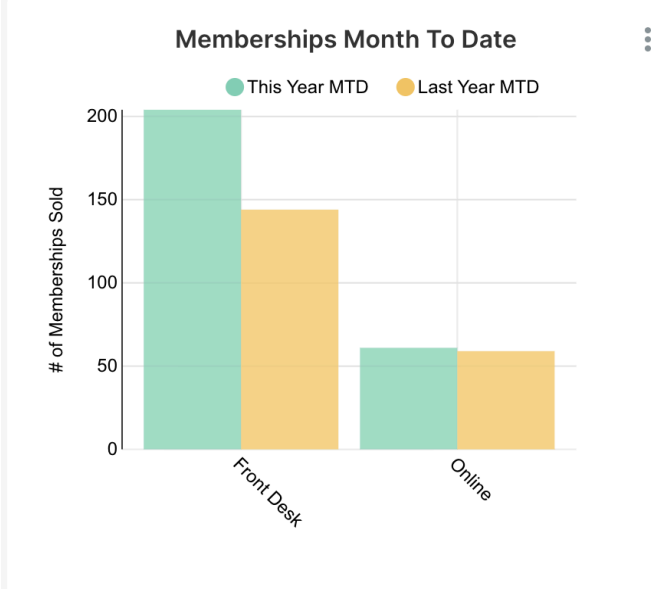
Figure shows the number of attendees that attended the museum through general admission successively. The green line tracks this year's data, whereas the orange line represents that same date 52 weeks ago,

The next chart we would like to display is the top 100 donors' most recent donation chart that we added to the donor dashboard. This chart has an interactive element where our client can hover over a specific bubble (high: >5 donations, medium: <=5 donations, low: >1 donation, derived from 75%ile, median, 25%ile) and see the name of the patron, their patron ID, the days since their last gift, their total amount donated, and the number of donations they have contributed. We took in feedback from our client to make the patron ID and donor name appear on the bubble together so they could easily identify the individual within their database. They will use this chart to look into people who have donated a large sum of money but haven't done so in a while. They can then reach back out to them to increase donor retention.



The figure above displays the days since a patron has donated to the museum, with each bubble representing more detailed information about the donations themselves.

Lastly, we want to feature some of the work we did for the membership dashboard. First, we developed a chart that shows how many memberships are being sold at the front desk versus online for both this year and last year. This shows the staff which sales method is doing better comparatively. As seen, front desk sales are doing much better this year compared to online sales. Overall, membership sales are up from last year taking into account both online and front desk sales.



This figure displays the number of memberships sold at the front desk and online with the green bar displaying this year's data and the orange bar displaying last year's.

Additionally, we displayed a chart that shows the proportion of museum members who are also donors. Although a more simple visual, we felt it was important to emphasize where donors are truly coming from to the leadership. The interactive feature allows users to hover over each segment to view total counts and percentage breakdowns. As of April 16, from checking the chart in the dashboard we learned that the museum has approximately 5,780 total members, of whom 552 are also donors, representing 9.56% of the total membership.

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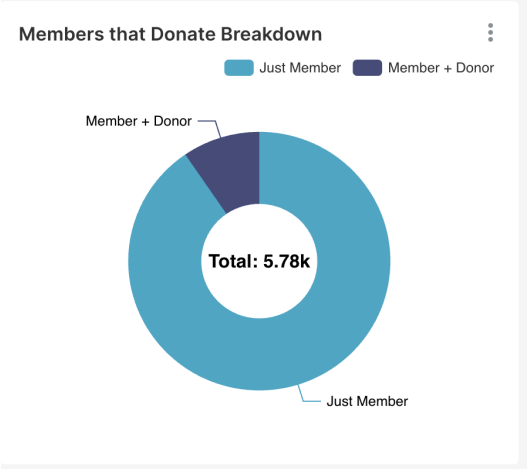


This figure shows the breakdown of members that actually donate, with the blue underscoring just members, and the purple representing members that are also donors.

Handoff & Documentation

To ensure a smooth handoff, we have created a “Data Refresh & Troubleshooting” guide as well as a “Project Documentation” document. The “Data Refresh & Troubleshooting” guide includes step-by-step instructions on how to refresh dashboard data. It also details how to fix common issues in Apache Superset, such as missing data or a dashboard not loading. It is designed for staff and future users to maintain dashboards without technical difficulties, regardless of background experience in dashboard usage. The “Project Documentation” document includes our project overview, objectives, and breaks down each visualization in each of the three dashboards, listing the type of chart, description, and purpose for their reference. An example of what this table looks like is below:

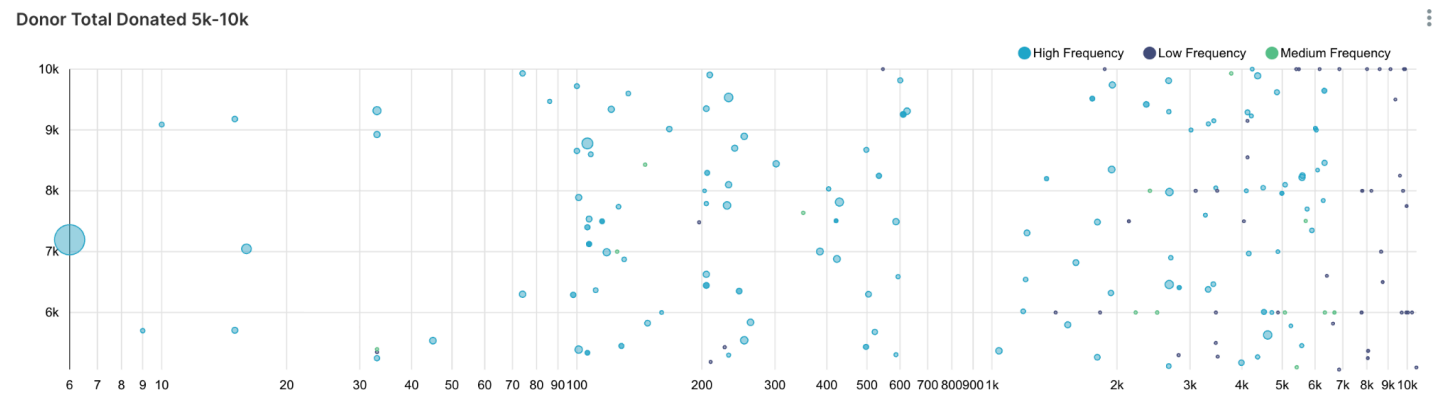
Visualization	Type	Description	Purpose
Cumulative Museum Attendance Vs. 52 Weeks Ago	Line Chart	This chart shows how the attendance changes compared to the previous year.	-Track growth over time -Identify trends and patterns -Compare performance across different periods -Support decision making

Above is an example of information we included in our user documentation guide for a specific chart

We also included information on how to access the dashboard and future work we'd like to see that could guide the direction of these dashboards in the long term. This document is designed to give the staff and future users in depth context for our project as they are using and improving the dashboards. We will be presenting these two documents, as well as walking through each dashboard in depth, to our client during our handoff meeting next week.

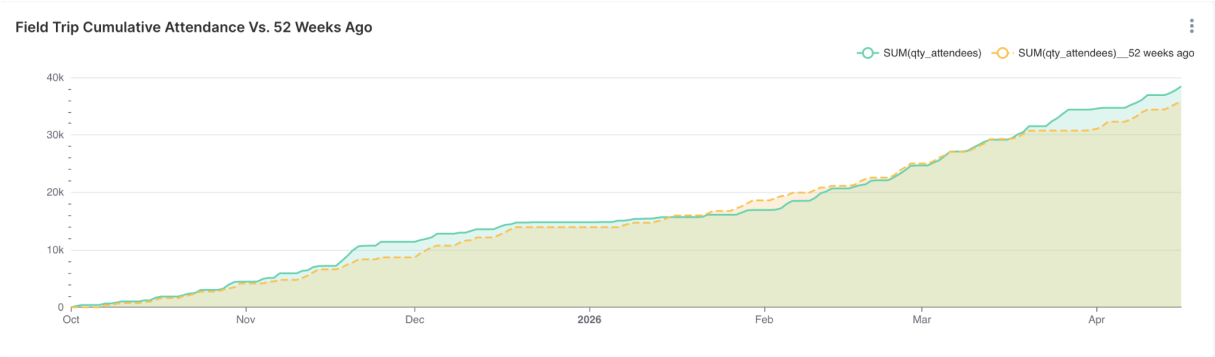
Appendix

1. Donor Total Donated 5k-10k



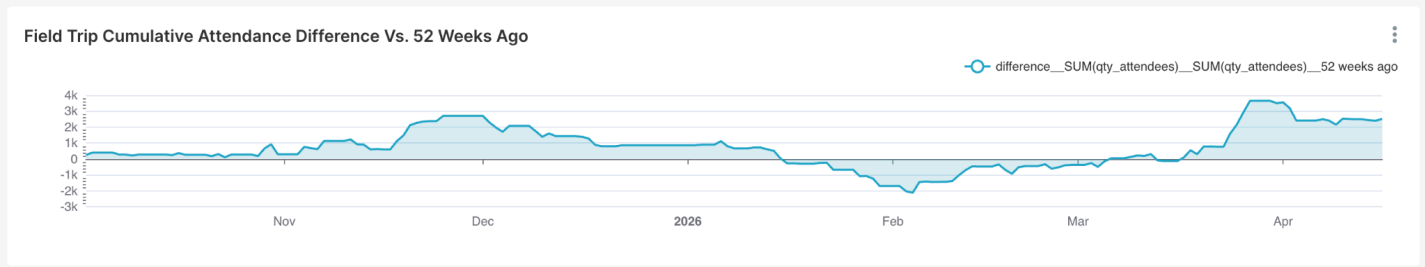
The figure above further segments the Top 100 Donors Most Recent Donation graph shown in the report. This one only includes donors whose largest gift was between $5,000 and $10,000

2. Field Trip Cumulative Attendance Vs. 52 weeks Ago



The figure above tracks cumulative attendance for each date for field trips, with the green line displaying the number of attendees for field trips this year, and the orange line indicating the number of attendees on field trips for the same date 52 weeks prior.

3. Field Trip Cumulative Attendance Difference Vs. 52 Weeks Ago



This chart exhibits the difference in cumulative attendance from last year to this year. We made this to show the sheer increase/decrease of attendees as it may be difficult to grasp the number when just looking at the cumulative chart.